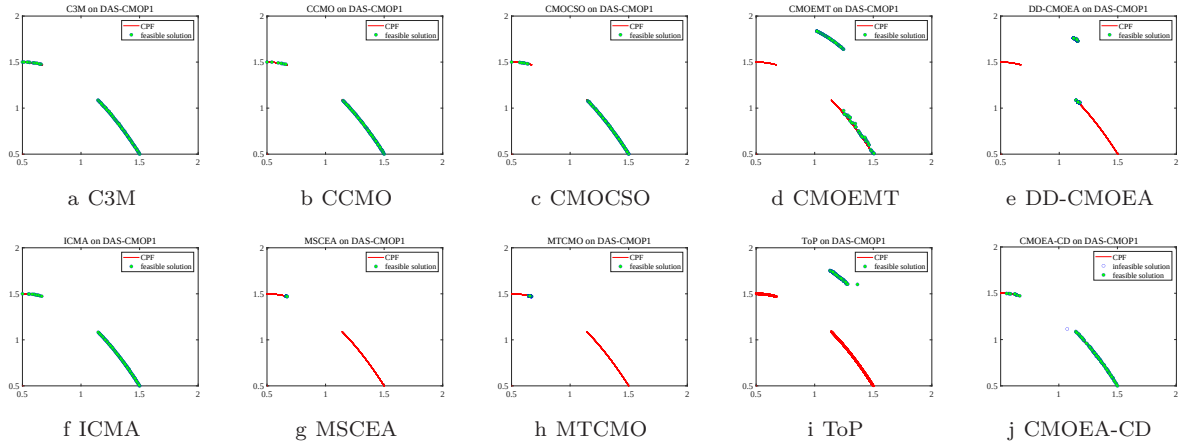
Constraint-Pareto Dominance and Diversity Enhancement Strategy based Evolutionary Algorithm for Solving Constrained Multiobjective Optimization Problems
Supplemental document

Table S - I: Parameter configurations of the test problems

Problems	M	D	parameter
DAS-CMOP1, DAS-CMOP2, DAS-CMOP3, DAS-CMOP4, DAS-CMOP5, DAS-CMOP6	2	30	-
DAS-CMOP7, DAS-CMOP8, DAS-CMOP9	3	30	-
LIR-CMOP1, LIR-CMOP2, LIR-CMOP3, LIR-CMOP4, LIR-CMOP5, LIR-CMOP6, LIR-CMOP7, LIR-CMOP8, LIR-CMOP9, LIR-CMOP10, LIR-CMOP11, LIR-CMOP12	2	30	-
LIR-CMOP13, LIR-CMOP14	3	30	-
MW1, MW2, MW3, MW5, MW6, MW7, MW9, MW10, MW11, MW12, MW13	2	15	-
MW4, MW8, MW14	3	15	-
DOC1	2	6	-
DOC2	2	16	-
DOC3	2	10	-
DOC4, DOC5	2	8	-
DOC6, DOC7	2	11	-
DOC8	3	10	-
DOC9	3	11	-
eq-DTLZ1, eq-IDTLZ1	3	7	k=3, p=1
eq-DTLZ2, eq-DTLZ3, eq-DTLZ4, eq-IDTLZ2	3	12	k=3, p=1

Table S - II: Parameter settings for comparison algorithms and CMOEA-CD

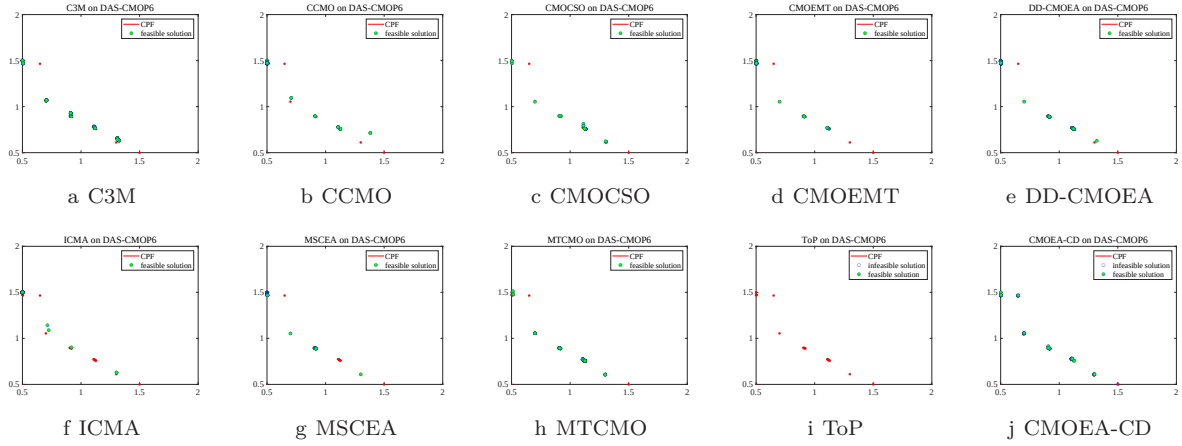
Algorithm	Character	Parameter	Operator	Operator parameter	Environmental selection method
C3M	Three-stage	-	DE	$CR = 1$, $F = 0.5$, $P_m = 1/D$, $\eta_m = 20$	Crowding distance based (SPEA2)
CCMO	Dual-population	-	DE	$CR = 1$, $F = 0.5$, $P_m = 1/D$, $\eta_m = 20$	Crowding distance based (SPEA2)
CMOCSO	Dual-population	-	CSO	$P_c = 1$, $\eta_c = 20$, $P_m = 1/D$, $\eta_m = 20$	Crowding distance based (SPEA2)
CMOEMT	Dual-stage and three-population	$\delta = 0.9$ $nr = 2$	GA DE	$P_c = 1$, $\eta_c = 20$, $P_m = 1/D$, $\eta_m = 20$ $CR = 1$, $F = 0.5$, $P_m = 1/D$, $\eta_m = 20$	Both crowding distance based (SPEA2) and reference vectors based
DD-CMOEA	Dual-stage and dual-population	$lastgen = 20$ $threshold = 1 \times 10^{-3}$	GA	$P_c = 1$, $\eta_c = 20$, $P_m = 1/D$, $\eta_m = 20$	Crowding distance based (SPEA2)
ICMA	Dual-population	-	DE DE	$CR = 1$, $F = 0.5$, $P_m = 1/D$, $\eta_m = 20$ $CR = 0.5$, $F = 0.5$, $P_m = 0.75/D$, $\eta_m = 0.75$	Both crowding distance based (SPEA2) and reference vectors based (PREA)
MSCEA	Three-stage and dual-population	$cp = 5$	GA	$P_c = 1$, $\eta_c = 20$, $P_m = 1/D$, $\eta_m = 20$	Crowding distance based (SPEA2)
MTCMO	Dual-population	-	GA	$P_c = 1$, $\eta_c = 20$, $P_m = 1/D$, $\eta_m = 20$	Crowding distance based (SPEA2)
ToP	Dual-stage	-	DE	$CR = 1$, $F = 0.5$, $P_m = 1/D$, $\eta_m = 20$	Crowding distance based (NSGA-II)
CMOEA-CD	Three-population	-	GA DE	$P_c = 1$, $\eta_c = 20$, $P_m = 1/D$, $\eta_m = 1$ $CR = 1$, $F = 0.5$, $P_m = 1/D$, $\eta_m = 1$	Both crowding distance based (SPEA2) and reference vectors based



a C3M b CCMO c CMOCSO d CMOEMT e DD-CMOEA

f ICMA g MSCEA h MTCMO i ToP j CMOEA-CD

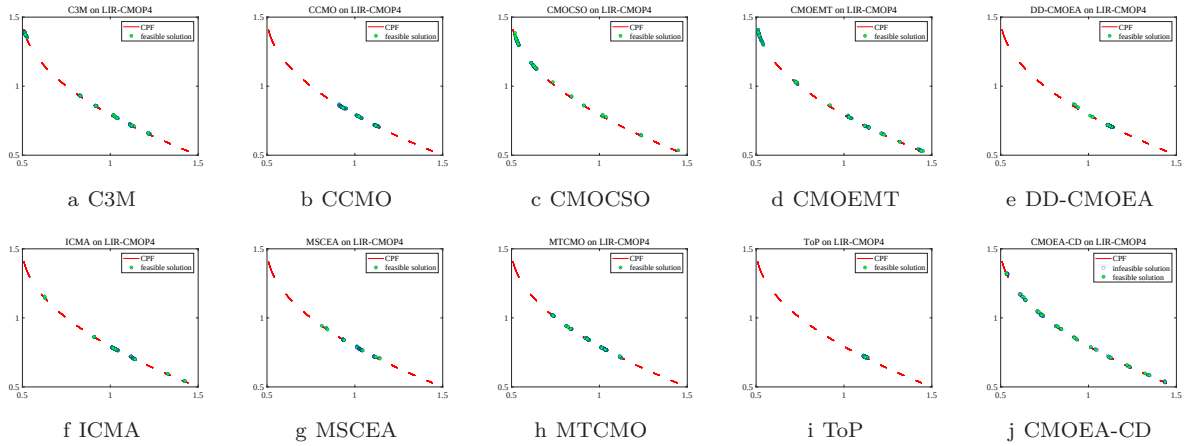
Fig. S - 1: Final results with median IGD value obtained by the comparison algorithms and CMOEA-CD on DAS-CMOP1



a C3M b CCMO c CMOCSO d CMOEMT e DD-CMOEA

f ICMA g MSCEA h MTCMO i ToP j CMOEA-CD

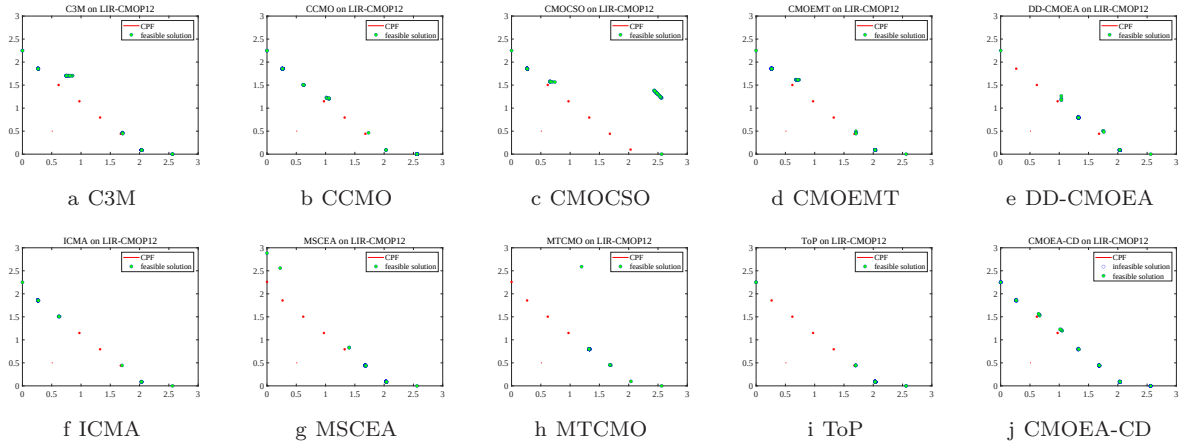
Fig. S - 2: Final results with median IGD value obtained by the comparison algorithms and CMOEA-CD on DAS-CMOP6



a C3M b CCMO c CMOCSO d CMOEMT e DD-CMOEA

f ICMA g MSCEA h MTCMO i ToP j CMOEA-CD

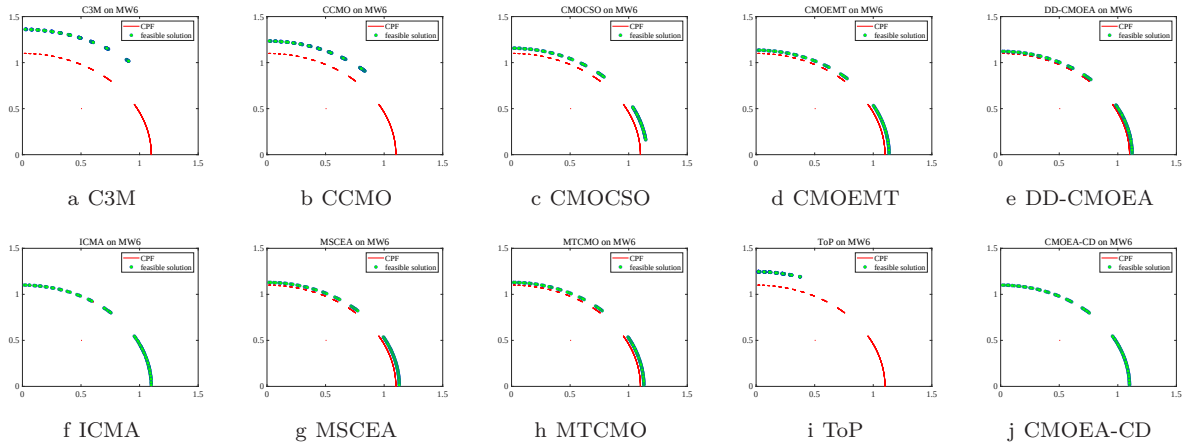
Fig. S - 3: Final results with median IGD value obtained by the comparison algorithms and CMOEA-CD on LIR-CMOP4



a C3M b CCMO c CMOCSO d CMOEMT e DD-CMOEA

f ICMA g MSCEA h MTCMO i ToP j CMOEA-CD

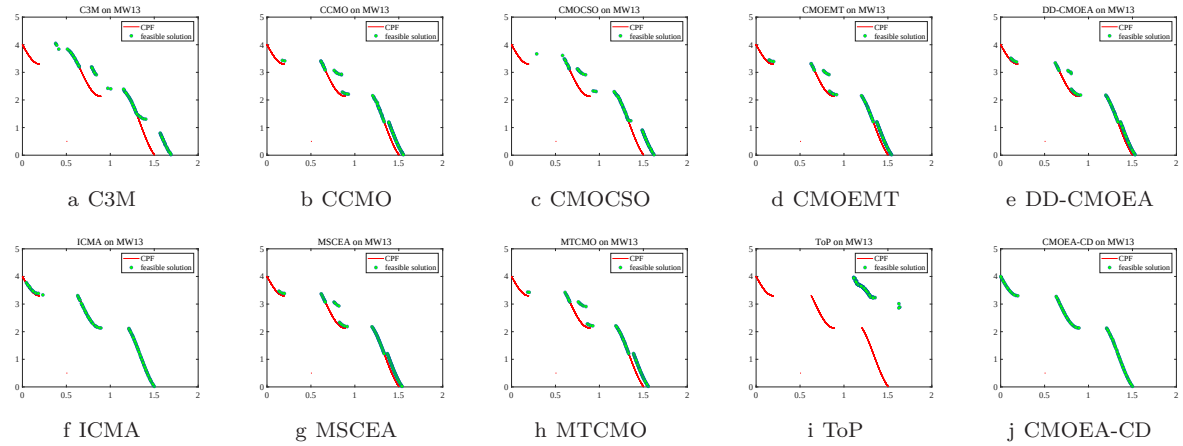
Fig. S - 4: Final results with median IGD value obtained by the comparison algorithms and CMOEA-CD on LIR-CMOP12



a C3M b CCMO c CMOCSO d CMOEMT e DD-CMOEA

f ICMA g MSCEA h MTCMO i ToP j CMOEA-CD

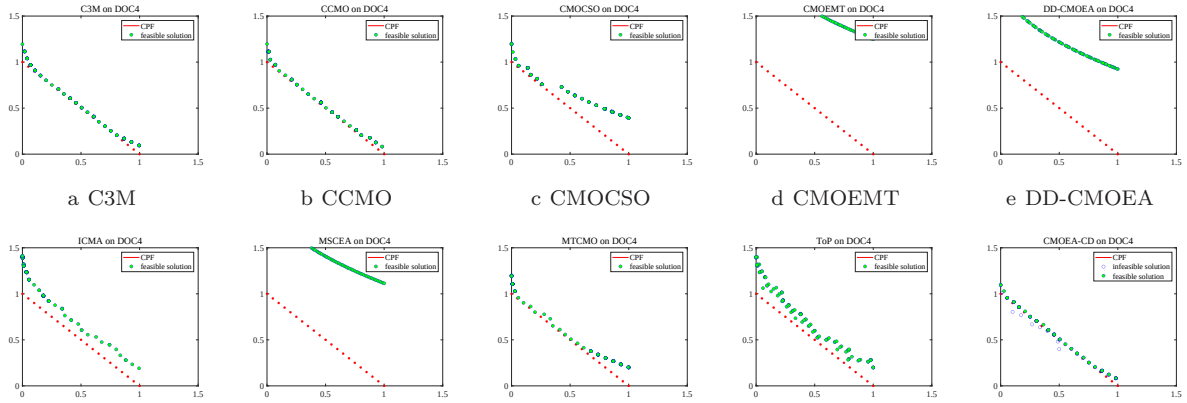
Fig. S - 5: Final results with median IGD value obtained by the comparison algorithms and CMOEA-CD on MW6



a C3M b CCMO c CMOCSO d CMOEMT e DD-CMOEA

f ICMA g MSCEA h MTCMO i ToP j CMOEA-CD

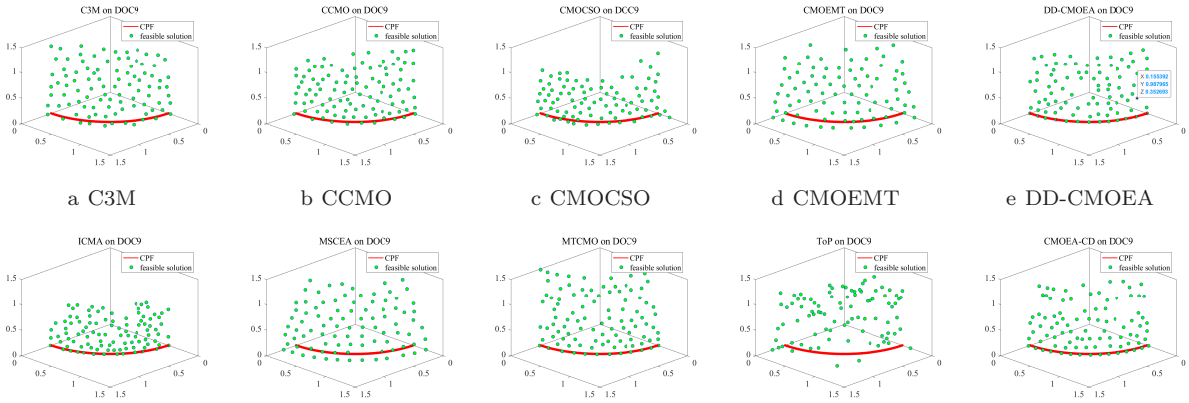
Fig. S - 6: Final results with median IGD value obtained by the comparison algorithms and CMOEA-CD on MW13



a C3M b CCMO c CMOCSO d CMOEMT e DD-CMOEA

f ICMA g MSCEA h MTCMO i ToP j CMOEA-CD

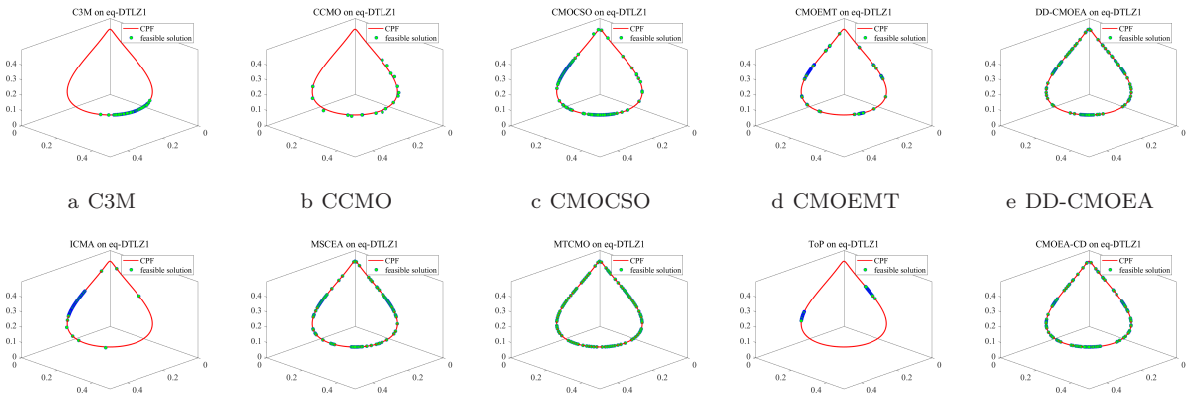
Fig. S - 7: Final results with median IGD value obtained by the comparison algorithms and CMOEA-CD on DOC4



a C3M b CCMO c CMOCSO d CMOEMT e DD-CMOEA

f ICMA g MSCEA h MTCMO i ToP j CMOEA-CD

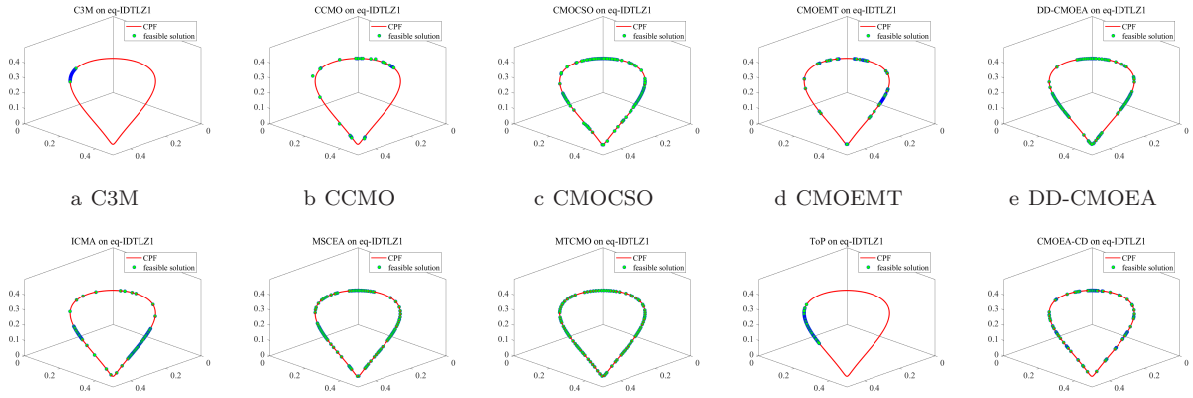
Fig. S - 8: Final results with median IGD value obtained by the comparison algorithms and CMOEA-CD on DOC9



a C3M b CCMO c CMOCSO d CMOEMT e DD-CMOEA

f ICMA g MSCEA h MTCMO i ToP j CMOEA-CD

Fig. S - 9: Final results with median IGD value obtained by the comparison algorithms and CMOEA-CD on eq-DTLZ1



a C3M b CCMO c CMOCSO d CMOEMT e DD-CMOEA

f ICMA g MSCEA h MTCMO i ToP j CMOEA-CD

Fig. S - 10: Final results with median IGD value obtained by the comparison algorithms and CMOEA-CD on eq-IDTLZ1

Table S - III: Statistical results of mean HV values obtained by comparison algorithms and CMOEA-CD on DAS-CMOP, LIR-CMOP, MW, DOC, and equality DTLZ benchmark test functions

Problems		C3M	CCMO	CMOCSO	CMOEMT	DD-CMOEA	ICMA	MSCEA	MTCMO	ToP	CMOEA-CD
DASCMOP1	mean	2.1161e-01	2.1115e-01	2.1123e-01	8.6714e-02	3.6740e-02	2.1207e-01	3.5900e-02	1.4070e-02	8.6650e-03	2.1143e-01
	std	1.16e-03(=)	8.81e-04(=)	1.29e-03(=)	8.55e-02(−)	2.80e-02(−)	4.91e-04(+)	2.57e-02(−)	1.13e-02(−)	2.47e-02(−)	7.43e-04
DASCMOP2	mean	3.5458e-01	3.5413e-01	3.5498e-01	3.1860e-01	2.7569e-01	3.5527e-01	2.6919e-01	2.6501e-01	6.6463e-02	3.5434e-01
	std	1.85e-04(=)	3.11e-04(=)	1.08e-04(+)	3.93e-02(−)	3.29e-03(−)	5.94e-05(+)	2.88e-03(−)	3.45e-03(−)	8.65e-02(−)	1.68e-04
DASCMOP3	mean	2.9742e-01	3.0814e-01	3.1145e-01	2.7526e-01	2.2244e-01	3.1175e-01	2.2609e-01	2.2199e-01	3.5960e-02	3.1099e-01
	std	2.48e-02(=)	1.10e-02(=)	6.26e-04(=)	4.69e-02(=)	1.62e-02(−)	5.28e-04(=)	1.54e-02(−)	1.37e-02(−)	5.02e-02(−)	6.59e-04
DASCMOP4	mean	1.9610e-01	1.7551e-01	1.9642e-01	1.9839e-01	2.0210e-01	1.5178e-01	2.0400e-01	2.0364e-01	0.0000e+00	2.0362e-01
	std	1.31e-02(−)	5.59e-02(−)	4.36e-03(−)	9.71e-03(=)	3.55e-03(−)	5.24e-02(−)	2.45e-04(+)	2.11e-03(=)	0.00e+00(−)	4.14e-04
DASCMOP5	mean	3.4587e-01	3.2162e-01	3.4769e-01	3.5017e-01	3.5142e-01	3.2070e-01	3.5143e-01	3.5158e-01	0.0000e+00	3.5091e-01
	std	2.47e-03(−)	6.66e-02(−)	7.61e-04(−)	1.11e-03(=)	1.25e-04(+)	7.57e-02(=)	1.49e-04(+)	8.14e-05(+)	0.00e+00(−)	3.25e-04
DASCMOP6	mean	2.8621e-01	2.7662e-01	3.0959e-01	3.0027e-01	3.0223e-01	2.0883e-01	2.9706e-01	2.0141e-01	0.0000e+00	3.1200e-01
	std	7.49e-02(−)	8.66e-02(−)	1.10e-03(−)	1.32e-02(−)	2.06e-02(=)	1.09e-01(−)	3.42e-02(=)	1.02e-01(−)	0.00e+00(−)	3.20e-04
DASCMOP7	mean	2.5589e-01	2.5391e-01	2.7922e-01	2.8516e-01	2.8802e-01	2.7580e-01	2.8754e-01	2.8848e-01	0.0000e+00	2.8683e-01
	std	4.56e-02(−)	6.73e-02(−)	1.40e-03(−)	7.32e-03(=)	3.07e-04(+)	2.70e-02(−)	3.88e-04(+)	3.79e-04(+)	0.00e+00(−)	4.94e-04
DASCMOP8	mean	1.7582e-01	1.7887e-01	2.0019e-01	2.0385e-01	2.0672e-01	2.0491e-01	2.0649e-01	2.0731e-01	0.0000e+00	2.0648e-01
	std	3.85e-02(−)	5.22e-02(−)	1.17e-03(−)	7.76e-03(−)	4.75e-04(=)	6.55e-03(−)	4.27e-04(=)	4.38e-04(+)	0.00e+00(−)	4.74e-04
DASCMOP9	mean	2.0185e-01	2.0326e-01	2.0459e-01	1.7883e-01	1.3076e-01	2.0675e-01	1.4217e-01	1.3864e-01	6.7509e-02	2.0482e-01
	std	1.22e-02(−)	6.73e-04(−)	6.91e-04(=)	3.16e-02(−)	1.24e-02(−)	2.59e-04(+)	1.10e-02(−)	1.02e-02(−)	3.29e-02(−)	4.40e-04
LIRCMOP1	mean	1.4650e-01	1.6247e-01	2.2022e-01	2.1125e-01	1.6445e-01	2.1026e-01	1.6810e-01	1.6630e-01	1.0869e-01	2.2476e-01
	std	2.82e-02(−)	2.89e-02(−)	9.79e-03(−)	2.44e-02(−)	1.64e-02(−)	1.53e-02(−)	1.08e-02(−)	6.04e-03(−)	9.87e-03(−)	2.70e-03
LIRCMOP2	mean	2.6302e-01	3.0145e-01	3.4872e-01	3.4882e-01	2.8444e-01	3.3894e-01	2.8666e-01	2.9022e-01	2.1669e-01	3.4681e-01
	std	3.14e-02(−)	3.03e-02(−)	8.42e-03(=)	1.68e-02(=)	1.79e-02(−)	1.02e-02(−)	1.10e-02(−)	8.71e-03(−)	1.32e-02(−)	2.34e-03
LIRCMOP3	mean	1.3173e-01	1.4720e-01	1.9074e-01	1.9497e-01	1.3588e-01	1.8061e-01	1.5189e-01	1.4817e-01	9.1739e-02	1.9826e-01
	std	2.88e-02(−)	2.20e-02(−)	9.82e-03(−)	6.27e-03(−)	1.84e-02(−)	1.26e-02(−)	1.06e-02(−)	9.17e-03(−)	5.89e-03(−)	3.91e-03
LIRCMOP4	mean	2.3664e-01	2.4875e-01	2.9902e-01	2.9941e-01	2.3585e-01	2.7682e-01	2.5135e-01	2.4847e-01	1.8262e-01	3.0394e-01
	std	3.39e-02(−)	3.04e-02(−)	9.25e-03(−)	2.14e-02(=)	2.38e-02(−)	1.63e-02(−)	8.63e-03(−)	1.45e-02(−)	1.10e-02(−)	4.42e-03
LIRCMOP5	mean	2.7729e-01	2.4903e-01	2.8794e-01	1.9130e-01	1.5314e-01	1.3855e-01	3.3774e-02	5.5673e-02	0.0000e+00	2.8603e-01
	std	8.81e-03(−)	8.52e-02(−)	7.78e-03(+)	8.31e-02(−)	2.23e-02(−)	1.23e-01(−)	6.29e-02(−)	7.48e-02(−)	0.00e+00(−)	2.12e-03
LIRCMOP6	mean	1.8189e-01	1.7942e-01	1.9206e-01	1.2710e-01	1.0920e-01	1.0084e-01	2.3312e-02	3.6242e-02	0.0000e+00	1.9336e-01
	std	3.26e-02(−)	3.38e-02(−)	8.61e-03(−)	3.33e-02(−)	1.44e-02(−)	8.15e-02(−)	4.28e-02(−)	4.86e-02(−)	0.00e+00(−)	7.30e-04
LIRCMOP7	mean	2.9137e-01	2.9223e-01	2.5658e-01	2.6796e-01	2.4807e-01	2.8796e-01	2.4667e-01	2.5148e-01	6.0491e-02	2.8510e-01
	std	7.17e-03(+)	3.84e-03(+)	1.18e-02(−)	1.15e-02(−)	1.06e-02(−)	9.37e-03(+)	7.63e-03(−)	7.56e-03(−)	1.12e-01(−)	1.28e-02
LIRCMOP8	mean	2.9196e-01	2.9304e-01	2.6882e-01	2.6113e-01	2.3568e-01	2.8920e-01	2.3504e-01	2.3211e-01	1.5210e-02	2.8869e-01
	std	6.43e-03(+)	8.63e-04(+)	2.46e-02(−)	1.63e-02(−)	1.27e-02(−)	1.18e-02(+)	9.87e-03(−)	1.16e-02(−)	5.82e-02(−)	9.52e-03
LIRCMOP9	mean	4.2866e-01	4.5432e-01	4.0658e-01	4.1294e-01	3.7783e-01	4.7378e-01	3.1652e-01	2.0202e-01	2.8933e-01	5.2881e-01
	std	5.17e-02(−)	3.51e-02(−)	7.36e-02(−)	6.92e-02(−)	7.31e-02(−)	3.07e-02(−)	8.94e-02(−)	7.16e-02(−)	7.77e-02(−)	8.36e-03
LIRCMOP10	mean	6.3517e-01	6.9545e-01	6.4619e-01	5.8164e-01	6.1804e-01	6.7748e-01	4.9260e-01	2.5236e-01	4.7216e-01	6.9610e-01
	std	4.72e-02(−)	6.24e-03(=)	3.84e-02(−)	4.68e-02(−)	2.99e-02(−)	3.77e-02(=)	1.56e-01(−)	2.05e-01(−)	8.66e-02(−)	5.80e-03
LIRCMOP11	mean	6.5505e-01	6.8806e-01	6.5765e-01	6.3155e-01	6.6361e-01	6.7266e-01	5.7975e-01	4.0129e-01	4.1322e-01	6.8983e-01
	std	4.18e-02(−)	4.12e-03(=)	3.96e-02(−)	4.78e-02(−)	2.10e-02(−)	3.44e-02(=)	1.08e-01(−)	1.35e-01(−)	8.01e-02(−)	3.24e-03
LIRCMOP12	mean	5.7729e-01	5.9486e-01	5.2586e-01	5.6766e-01	5.3940e-01	5.7558e-01	4.9774e-01	4.5061e-01	4.6002e-01	6.1042e-01
	std	4.55e-02(=)	1.60e-02(−)	6.51e-02(−)	2.52e-02(−)	3.51e-02(−)	2.06e-02(−)	5.18e-02(−)	4.09e-02(−)	3.61e-02(−)	7.79e-03
LIRCMOP13	mean	4.9140e-01	5.0664e-01	5.4459e-01	5.3452e-01	5.5214e-01	5.4547e-01	1.3878e-04	1.1878e-04	3.7182e-03	5.5780e-01
	std	3.38e-02(−)	2.48e-03(−)	1.88e-03(−)	5.69e-03(−)	1.43e-03(−)	1.33e-03(−)	1.52e-04(−)	1.40e-04(−)	1.39e-02(−)	1.63e-03
LIRCMOP14	mean	5.3321e-01	5.3766e-01	5.4853e-01	5.4752e-01	5.5147e-01	5.5308e-01	5.1738e-04	5.3298e-04	3.2245e-03	5.5319e-01
	std	1.38e-02(−)	2.03e-03(−)	1.73e-03(−)	2.17e-03(−)	1.28e-03(−)	1.00e-03(=)	2.87e-04(−)	3.14e-04(−)	1.63e-02(−)	1.56e-03
MW1	mean	3.6606e-01	4.8233e-01	4.9000e-01	4.8963e-01	4.9005e-01	4.8848e-01	4.9008e-01	4.8973e-01	0.0000e+00	4.9010e-01
	std	1.71e-01(−)	1.40e-02(−)	3.96e-05(−)	1.86e-04(−)	4.09e-05(−)	3.23e-04(−)	2.88e-05(−)	2.01e-03(−)	0.00e+00(=)	1.83e-05
MW2	mean	4.5727e-01	4.3323e-01	5.1061e-01	5.4752e-01	5.5055e-01	5.7998e-01	5.5690e-01	5.5166e-01	4.1508e-01	5.8240e-01
	std	6.87e-02(−)	5.30e-02(−)	4.03e-02(−)	1.38e-02(−)	1.35e-02(−)	1.41e-03(−)	1.13e-02(−)	1.64e-02(−)	1.12e-01(−)	4.49e-05
MW3	mean	5.4263e-01	5.4331e-01	5.4422e-01	5.4408e-01	5.4460e-01	5.4495e-01	5.4440e-01	5.4455e-01	1.7739e-01	5.4411e-01
	std	5.50e-04(−)	2.16e-04(−)	3.48e-04(=)	5.27e-04(=)	7.50e-04(=)	3.03e-04(+)	5.37e-04(=)	3.80e-04(=)	1.95e-01(−)	6.36e-04
MW4	mean	8.1402e-01	7.9027e-01	8.3951e-01	8.3376e-01	8.3998e-01	8.3424e-01	8.4023e-01	8.4004e-01	9.1495e-02	8.4155e-01
	std	4.22e-02(−)	1.16e-01(−)	6.49e-04(−)	2.64e-03(−)	4.21e-04(−)	8.33e-04(−)	3.95e-04(−)	5.54e-04(−)	0.00e+00(=)	4.82e-04
MW5	mean	1.8982e-01	2.4927e-01	3.2223e-01	3.2357e-01	3.2371e-01	3.2095e-01	3.1905e-01	3.2109e-01	0.0000e+00	3.2425e-01
	std	1.02e-01(−)	1.01e-01(−)	7.87e-04(−)	5.83e-04(−)	3.21e-03(=)	1.01e-03(−)	2.80e-02(−)	1.87e-02(−)	0.00e+00(−)	1.71e-04
MW6	mean	1.2584e-01	1.5584e-01	1.9346e-01	2.8166e-01	3.0074e-01	3.2687e-01	3.0316e-01	2.9954e-01	8.4285e-02	3.2850e-01
	std	5.15e-02(−)	7.01e-02(−)	8.60e-02(−)	4.78e-02(−)	1.91e-02(−)	5.73e-04(−)	1.25e-02(−)	2.18e-02(−)	8.86e-02(−)	1.98e-05
MW7	mean	4.1142e-01	4.1195e-01	4.1212e-01	4.1187e-01	4.1216e-01	4.1141e-01	4.1185e-01	4.1214e-01	3.6307e-01	4.1167e-01
	std	3.90e-04(=)	1.60e-04(+)	3.99e-04(+)	4.93e-04(=)	6.51e-04(+)	5.44e-04(=)	6.21e-04(=)	6.05e-04(+)	6.31e-02(−)	4.99e-04
MW8	mean	3.8093e-01	3.3960e-01	4.2925e-01	5.2429e-01	5.2774e-01	5.1488e-01	5.3097e-01	5.3079e-01	1.1012e-01	5.5366e-01
	std	8.59e-02(−)	8.21e-02(−)	8.67e-02(−)	3.04e-02(−)	1.78e-02(−)	9.98e-03(−)	1.54e-02(−)	1.47e-02(−)	1.18e-01(−)	7.30e-04
MW9	mean	2.7568e-01	2.8787e-01	3.9739e-01	3.9706e-01	3.8630e-01	3.9593e-01	3.9498e-01	3.8402e-01	0.0000e+00	3.9646e-01
	std	1.70e-01(−)	1.77e-01(−)	1.58e-03(+)	2.44e-03(=)	7.30e-02(−)	1.78e-03(=)	3.18e-03(−)	7.26e-02(−)	0.00e+00(−)	2.10e-03
MW10	mean	2.4669e-01	2.3061e-01	2.8751e-01	4.0767e-01	4.0559e-01	4.5270e-01	4.3018e-01	4.1579e-01	1.7449e-01	4.5495e-01
	std	7.26e-02(−)	1.03e-01(−)	1.20e-01(−)	5.27e-02(−)	3.46e-02(−)	2.56e-03(−)	1.62e-02(−)	2.43e-02(−)	0.00e+00(=)	7.98e-05
MW11	mean	4.4675e-01	4.4653e-01	4.4391e-01	4.4746e-01	4.4789e-01	4.4730e-01	4.4763e-01	4.4785e-01	2.7863e-01	4.4724e-01
	std	2.73e-04(−)	3.47e-04(−)	2.07e-03(−)	2.04e-04(=)	1.26e-04(+)	3.04e-04(=)	2.55e-04(+)	1.32e-04(+)	3.26e-02(−)	2.14e-04
MW12	mean	3.6843e-01	4.4178e-01	6.0330e-01	6.0400e-01	5.8515e-01	6.0444e-01	6.0489e-01	6.0509e-01	1.6952e-02	6.0423e-01
	std	2.71e-01(−)	2.52e-01(−)	2.50e-04(−)	3.56e-04(=)	1.07e-01(−)	2.12e-04(=)	1.90e-04(+)	3.36e-04(+)	2.03e-02(−)	3.20e-04
MW13	mean	3.4694e-01	3.4353e-01	3.8274e-01	4.3355e-01	4.5059e-01	4.7364e-01	4.4124e-01	4.4066e-01	2.3389e-01	4.7689e-01
	std	5.46e-02(−)	5.00e-02(−)	5.41e-02(−)	2.48e-02(−)	1.80e-02(−)	2.32e-03(−)	2.43e-02(−)	3.10e-02(−)	1.19e-01(−)	6.3e-05
MW14	mean	4.5361e-01	4.5850e-01	4.7032e-01	4.6797e-01	4.7154e-01	4.6407e-01	4.7248e-01	4.7210e-01	3.4117e-01	4.7452e-01
	std	6.82e-03(−)	4.49e-03(−)	1.39e-03(−)	2.07e-03(−)	1.83e-03(−)	2.41e-03(−)	1.53e-03(−)	1.95e-03(−)	1.64e-01(−)	1.33e-03
DOC1	mean	3.4527e-01	3.4461e-01	2.8492e-01	1.2401e-01	1.9854e-02	3.2545e-01	3.2359e-01	3.4249e-01	3.4454e-01	3.4615e-01
	std	4.32e-04(−)	7.54e-04(−)	8.28e-02(−)	1.34e-01(−)	6.21e-02(−)	7.70e-02(−)	3.29e-02(−)	5.85e-03(−)	3.83e-04(−)	4.68e-04
DOC2	mean	5.7844e-01	6.0247e-01	1.8635e-01	3.4178e-01	0.0000e+00	0.0000e+00	0.0000e+00	2.8409e-01	2.9036e-01	6.0832e-01
	std	8.47e-02(−)	8.44e-02(=)	0.00e+00(=)	2.19e-01(−)	0.00e+00(−)	0.00e+00(−)	0.00e+00(−)	1.84e-01(−)	1.07e-01(−)	1.61e-02
DOC3	mean	9.8160e-02	1.1034e-02	3.9345e-03	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	1.3193e-02
	std	1.53e-01(−)	6.04e-02(=)	1.89e-02(=)	0.00e+00(−)	0.00e+00(−)	0.00e+00(−)	0.00e+00(−)	0.00e+00(−)	0.00e+00(−)	6.60e-02
DOC4	mean	5.3992e-01	5.3631e-01	2.6472e-01	1.5286e-01	6.5955e-02	3.6435e-01	4.0701e-01	4.9878e-01	4.8917e-01	5.3510e-01
	std	4.76e-03(=)	4.01e-03(−)	1.04e-01(−)	1.49e-01(−)	1.27e-01(−)	1.35e-01(−)	1.64e-01(−)	5.64e-02(−)	6.92e-02(=)	5.46e-03
DOC5	mean	4.1010e-01	4.3572e-01	0.0000e+00	1.4491e-01	0.0000e+00	1.9868e-01	0.0000e+00	0.0000e+00	2.4180e-01	1.4228e-01
	std	1.79e-01(+)	1.16e-01(+)	0.00e+00(−)	1.64e-01(=)	0.00e+00(−)	2.14e-01(+)	0.00e+00(−)	0.00e+00(−)	2.05e-01(+)	2.04e-01
DOC6	mean	5.2859e-01	4.9094e-01	1.9479e-01	9.4199e-02	2.6587e-02	1.2193e-01	2.5405e-01	2.3561e-01	0.0000e+00	5.2915e-01
	std	1.39e-02(=)	4.52e-02(−)	1.59e-01(−)	1.44e-01(−)	7.58e-02(−)	1.86e-01(−)	1.94e-01(−)	1.90e-01(−)	0.00e+00(−)	7.53e-03
DOC7	mean	5.0892e-01	5.2297e-01	7.6673e-02	6.8964e-02	0.0000e+00	4.0157e-03	0.0000e+00	6.8316e-02	2.2459e-01	3.1184e-01
	std	9.73e-02(+)	4.15e-02(+)	1.18e-01(−)	1.07e-01(=)	0.00e+00(−)	2.13e-02(−)	0.00e+00(−)	1.31e-01(−)	1.76e-01(=)	2.31e-01
DOC8	mean	8.0274e-01	7.8851e-01	0.0000e+00	0.0000e+00	0.0000e+00	2.3440e-01	0.0000e+00	0.0000e+00	0.0000e+00	2.3237e-01
	std	3.49e-03(+)	6.72e-03(+)	0.00e+00(−)	0.00e+00(−)	0.00e+00(−)	2.39e-01(=)	0.00e+00(−)	0.00e+00(−)	0.00e+00(−)	2.78e-01
DOC9	mean	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00	0.0000e+00
	std	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)	0.00e+00(=)	0.00e+00
eq-DTLZ1	mean	1.9759e-01	3.9777e-01	4.8247e-01	4.1010e-01	4.8095e-01	4.1824e-01	4.8868e-01	4.9437e-01	1.9206e-01	4.8487e-01
	std	7.94e-02(−)	3.24e-02(−)	5.35e-03(=)	4.12e-02(−)	2.20e-02(=)	2.89e-02(−)	2.52e-03(+)	1.04e-03(+)	1.20e-01(−)	6.03e-03
eq-DTLZ2	mean	2.7447e-01	3.1888e-01	3.4402e-01	3.4053e-01	3.3959e-01	3.2384e-01	3.4717e-01	3.4782e-01	2.3255e-01	3.4298e-01
	std	5.43e-02(−)	6.23e-03(−)	1.15e-03(+)	2.49e-03(=)	4.17e-03(−)	1.20e-02(−)	3.87e-04(+)	1.61e-04(+)	6.07e-02(−)	1.68e-03
eq-DTLZ3	mean	7.9343e-02	1.6628e-01	3.2869e-01	2.7419e-01	3.1088e-01	2.0998e-01	3.3777e-01	3.4415e-01	8.9455e-02	3.3485e-01
	std	6.05e-02(−)	9.29e-02(−)	4.99e-03(−)	2.66e-02(−)	1.30e-02(−)	4.39e-02(−)	3.00e-03(+)	1.51e-03(+)	6.87e-02(−)	4.29e-03
eq-DTLZ4	mean	2.7585e-01	2.9253e-01	3.4034e-01	3.4230e-01	3.3532e-01	3.2819e-01	3.4679e-01	3.4777e-01	2.3961e-01	3.3873e-01
	std	2.82e-02(−)	1.75e-02(−)	2.08e-03(=)	2.82e-03(=)	5.51e-03(−)	1.06e-02(−)	6.26e-04(+)	1.37e-04(+)	7.31e-02(−)	3.86e-03
eq-DTLZ5	mean	4.0238e-02	8.9152e-02	1.1219e-01	9.0310e-02	1.0610e-01	1.0171e-01	1.1354e-01	1.1642e-01	3.6343e-02	1.1281e-01
	std	1.78e-02(−)	7.58e-03(−)	1.49e-03(=)	9.12e-03(−)	2.86e-03(−)	5.93e-03(−)	1.57e-03(+)	7.49e-04(+)	1.91e-02(−)	1.81e-03
eq-DTLZ6	mean	2.2849e-01	2.4927e-01	2.8836e-01	2.9155e-01	2.8167e-01	2.7091e-01	2.9224e-01	2.9318e-01	1.8024e-01	2.8672e-01
	std	3.05e-02(−)	1.08e-02(−)	1.43e-03(+)	9.68e-04(+)	4.46e-03(−)	1.20e-02(−)	6.57e-04(+)	2.48e-04(+)	5.56e-02(−)	2.07e-03
+/-/=		5/39/8	6/38/8	6/35/11	1/35/16	4/42/6	7/34/11	11/36/5	12/37/3	1/45/6	

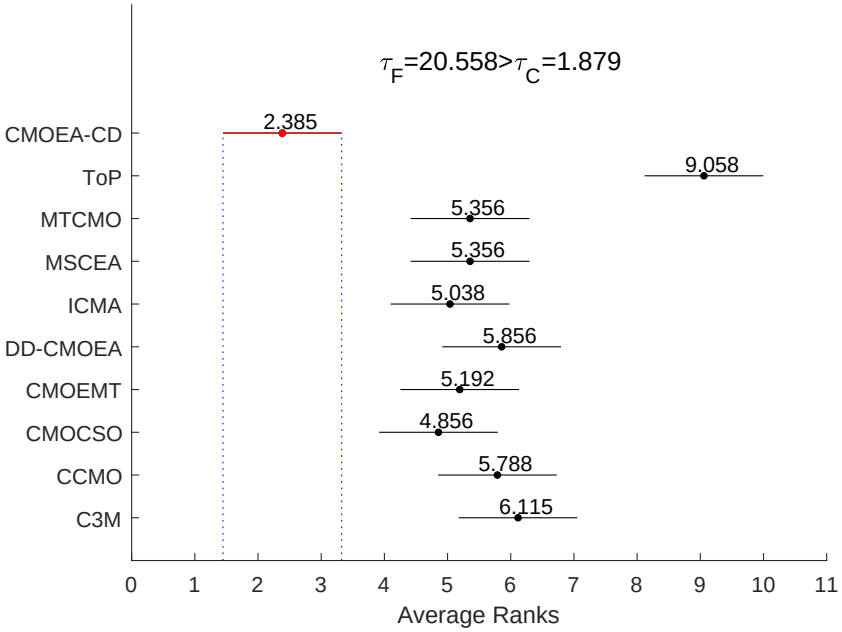


Fig. S - 11: The Friedman Test results of the HV value obtained by comparison algorithms and CMOEA-CD on DAS-CMOP, LIR-CMOP, MW, DOC, and equality DTLZ benchmark test functions.

Table S - IV: Ablation experimental results of HV values obtained by CMOEA-CD-WCPD, CMOEA-CD-WD, and CMOEA-CD on DAS-CMOP, LIR-CMOP, MW, DOC, equality DTLZ, and DC-DTLZ benchmark test functions

Problems	M	CMOEA-CD-WCPD	CMOEA-CD-WD	CMOEA-CD
DASCMOP1	2	2.1038e-01(9.12e-04)−	2.1006e-01(9.77e-04)−	2.1143e-01(7.43e-04)
DASCMOP2	2	3.5428e-01(1.56e-04)=	3.5314e-01(1.02e-03)−	3.5434e-01(1.68e-04)
DASCMOP3	2	3.1067e-01(9.91e-04)=	2.8364e-01(3.20e-02)−	3.1099e-01(6.59e-04)
DASCMOP4	2	2.0261e-01(2.75e-03)=	2.0327e-01(1.64e-03)=	2.0362e-01(4.14e-04)
DASCMOP5	2	3.5032e-01(5.88e-04)−	3.5085e-01(2.45e-04)=	3.5091e-01(3.25e-04)
DASCMOP6	2	3.1205e-01(3.05e-04)=	3.1173e-01(5.80e-04)=	3.1200e-01(3.20e-04)
DASCMOP7	3	2.8649e-01(5.56e-04)−	2.8659e-01(6.50e-04)=	2.8683e-01(4.94e-04)
DASCMOP8	3	2.0570e-01(5.52e-04)−	2.0542e-01(7.13e-04)−	2.0648e-01(4.74e-04)
DASCMOP9	3	2.0475e-01(4.06e-04)=	2.0347e-01(1.23e-03)−	2.0482e-01(4.40e-04)
LIRCMOP1	2	2.2167e-01(4.54e-03)−	1.4977e-01(1.69e-02)−	2.2476e-01(2.70e-03)
LIRCMOP2	2	3.4190e-01(4.18e-03)−	2.8457e-01(3.00e-02)−	3.4681e-01(2.34e-03)
LIRCMOP3	2	1.9521e-01(4.21e-03)−	1.2851e-01(1.79e-02)−	1.9826e-01(3.91e-03)
LIRCMOP4	2	2.9941e-01(7.26e-03)−	2.3477e-01(2.57e-02)−	3.0394e-01(4.42e-03)
LIRCMOP5	2	2.7414e-01(1.90e-02)−	2.8462e-01(3.78e-03)=	2.8603e-01(2.12e-03)
LIRCMOP6	2	1.8510e-01(1.01e-02)−	1.9016e-01(8.06e-03)−	1.9336e-01(7.30e-04)
LIRCMOP7	2	2.7750e-01(1.43e-02)−	2.7854e-01(1.42e-02)=	2.8510e-01(1.28e-02)
LIRCMOP8	2	2.8003e-01(1.26e-02)−	2.8292e-01(1.17e-02)−	2.8869e-01(9.52e-03)
LIRCMOP9	2	4.7723e-01(4.15e-02)−	5.0463e-01(3.19e-02)−	5.2881e-01(8.36e-03)
LIRCMOP10	2	6.7193e-01(3.65e-02)−	6.9392e-01(1.93e-02)−	6.9610e-01(5.80e-03)
LIRCMOP11	2	6.8160e-01(1.54e-02)=	6.9160e-01(1.57e-03)+	6.8983e-01(3.24e-03)
LIRCMOP12	2	5.8014e-01(2.66e-02)−	6.0054e-01(1.59e-02)=	6.1042e-01(7.79e-03)
LIRCMOP13	3	5.3444e-01(2.73e-03)−	5.3749e-01(2.28e-03)−	5.5780e-01(1.63e-03)
LIRCMOP14	3	5.4820e-01(2.06e-03)−	5.5092e-01(1.61e-03)−	5.5319e-01(1.56e-03)
MW1	2	4.8982e-01(5.7ee-05)−	4.8996e-01(3.8ee-05)−	4.9010e-01(1.83e-05)
MW2	2	5.8161e-01(2.04e-04)−	5.8196e-01(1.33e-04)−	5.8240e-01(4.49e-05)
MW3	2	5.4334e-01(6.57e-04)−	5.4396e-01(5.67e-04)=	5.4411e-01(6.36e-04)
MW4	3	8.3633e-01(1.03e-03)−	8.3753e-01(1.13e-03)−	8.4155e-01(4.82e-04)
MW5	2	3.2416e-01(2.02e-04)=	3.2431e-01(1.80e-04)=	3.2425e-01(1.71e-04)
MW6	2	3.2845e-01(2.11e-05)−	3.2848e-01(2.07e-05)−	3.2850e-01(1.98e-05)
MW7	2	4.1158e-01(3.56e-04)=	4.1204e-01(4.83e-04)+	4.1167e-01(4.99e-04)
MW8	3	5.4775e-01(3.89e-03)−	5.4826e-01(3.71e-03)−	5.5366e-01(7.30e-04)
MW9	2	3.9577e-01(2.97e-03)=	3.9743e-01(2.24e-03)=	3.9646e-01(2.10e-03)
MW10	2	4.5482e-01(8.88e-05)−	4.5473e-01(9.77e-05)−	4.5495e-01(7.98e-05)
MW11	2	4.4711e-01(2.42e-04)=	4.4704e-01(3.26e-04)−	4.4724e-01(2.14e-04)
MW12	2	6.0374e-01(3.75e-04)−	6.0418e-01(3.56e-04)=	6.0423e-01(3.20e-04)
MW13	2	4.7622e-01(2.00e-03)−	4.7688e-01(6.26e-05)=	4.7689e-01(6.3ee-05)
MW14	3	4.7197e-01(1.50e-03)−	4.6998e-01(1.94e-03)−	4.7452e-01(1.33e-03)
DOC1	2	3.4452e-01(9.14e-04)−	3.4620e-01(3.60e-04)=	3.4615e-01(4.68e-04)
DOC2	2	5.4710e-01(1.22e-01)−	5.8064e-01(1.01e-01)=	6.0832e-01(1.61e-02)
DOC3	2	3.2449e-03(1.65e-02)=	4.0282e-02(9.29e-02)=	1.3193e-02(6.60e-02)
DOC4	2	5.1501e-01(2.41e-02)−	5.3725e-01(6.35e-03)=	5.3510e-01(5.46e-03)
DOC5	2	4.7079e-01(1.15e-02)+	3.9674e-01(3.41e-02)+	1.4228e-01(2.04e-01)
DOC6	2	4.7640e-01(9.90e-02)−	5.2808e-01(4.41e-03)=	5.2915e-01(7.53e-03)
DOC7	2	9.2682e-02(1.85e-01)−	5.1997e-01(5.71e-03)+	3.1184e-01(2.31e-01)
DOC8	3	2.4025e-01(2.45e-01)=	5.9499e-01(2.74e-01)+	2.3237e-01(2.78e-01)
DOC9	3	0.0000e+00(0.00e+00)=	0.0000e+00(0.00e+00)=	0.0000e+00(0.00e+00)
eq-DTLZ1	3	4.3709e-01(2.38e-02)−	4.8138e-01(8.20e-03)=	4.8487e-01(6.03e-03)
eq-DTLZ2	3	3.3509e-01(5.22e-03)−	3.4318e-01(2.53e-03)=	3.4298e-01(1.68e-03)
eq-DTLZ3	3	2.8302e-01(2.08e-02)−	3.3515e-01(6.12e-03)=	3.3485e-01(4.29e-03)
eq-DTLZ4	3	3.3316e-01(3.78e-03)−	3.3857e-01(2.61e-03)=	3.3873e-01(3.86e-03)
eq-IDTLZ1	3	9.7291e-02(7.06e-03)−	1.1262e-01(2.35e-03)=	1.1281e-01(1.81e-03)
eq-IDTLZ2	3	2.7738e-01(3.93e-03)−	2.8754e-01(2.27e-03)=	2.8672e-01(2.07e-03)
+/-/=		1/38/13	5/23/24	